

Jalen Simpson

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EDUCATION

University of Louisville | Louisville, KY | Expected: December 2026

Degree: Bachelor of Science, Mechanical Engineering

Relevant Coursework: Machine Design I & II, Mechanical Vibrations, System Dynamics, Vehicle Dynamics and Handling

TECHNICAL SKILLS

CAD/Engineering: SolidWorks, AutoCAD, Catia, Inventor, Fusion 360, Autodesk Forma/ACC

Prototyping & Manufacturing: 3D printing, / Additive Manufacturing, Laser Systems, Mechanical Assembly

Analytics: Power BI, Power Query, Excel, ProEst, Quantity Takeoffs, Conceptual Estimating

EXPERIENCE

Toyota Motor North America | Georgetown, KY

Production Engineering Co-Op – Plant Engineering | May 2026–August 2026

- Assessed ~10 industrial heating and ventilation units at a manufacturing plant, identifying 3 high-priority assets and documenting mechanical, structural, operational, and fire-safety risks to support capital renewal planning.
- Developed 4 conceptual estimates totaling ~\$7M for facility and infrastructure projects using ProEst and Autodesk Forma/ACC, performing quantity takeoffs, scope development, and cost analysis for projects including utility, structural, and electrical upgrades.
- Evaluated HVAC alternatives across 11 North American plants and 4 regions using outside-air requirements, climate data, energy use, CO₂ emissions, and cost, identifying an alternative projected to save ~\$10K annually after the break-even year while supporting future engineering standards
- Built and automated a 4-KPI Power BI dashboard analyzing 800+ qualified engineering projects from 1,000+ records, automating the data-collection workflow and providing leadership visibility into process bottlenecks, performance gaps, and Kaizen opportunities.

Production Engineering Co-Op – Innovation and Technology | May 2025–August 2025

- Executed a 3-week technological trial by programming and operating robotic paint systems, supporting accelerated process testing and validation.
- Executed controlled paint trials and quality checks including paint mixing, viscosity testing, contamination control, and robotic component changeovers to maintain repeatable test conditions
- Co-authored a performance report for Toyota HQ Japan and presented findings to executives and engineering teams, informing evaluation of future automation initiatives

Production Engineering Co-Op – Operations | August 2024–December 2024

- Conducted field assessments of 15 North American storerooms using laser measurement tools and AutoCAD to document existing conditions and identify operational and layout gaps.
- Developed and presented standardized layout recommendations to Toyota leadership, translating field data into scalable facility-planning improvements

United Parcel Service (UPS) | Louisville, KY

December 2025–May 2026

Building and Systems Engineer – Operations Process Control

- Monitored building and conveyor systems across 3 facilities using SCADA and system data, identifying faults and coordinating corrective response with maintenance, management, and operations teams
- Performed system troubleshooting and reprogrammed building/conveyor controls to resolve equipment issues and adapt system performance to changing operational requirements.

GE Appliances/FirstBuild | Louisville, KY

November 2022 – May 2024

Engineering Assembly Technician/Student Worker

- Designed and implemented two assembly-process improvements in SolidWorks and CATIA, improving manufacturability and reducing process time.
- Used additive manufacturing and laser systems to prototype and fabricate three subassemblies and one final assembly to engineering specifications.